Table 1. Flow rates and normal boiling point ranges for products from processing of 10,000 BPSD of Poseidon Crude (58,018 kg/h). Design NBP range and flow rates are from CHEMCAD.

Product	Composition	Spec NBP Range, °C	Design NBP Range, °C	Flow Rate, BPSD	Flow Rate, kg/h
Light Ends / Tail Gas	C1 to C4	<15	-70 to 8	398	1,744
Light Naphtha	C5 to C6	15 to 75	16 to 110	1,106	5,347
Heavy Naphtha	C6 to C12	75 to 165	98 to 150	687	3,500
Kerosene	C6 to C16	165 to 200	145 to 243	1,289	6,935
GO/D	C12 to C20	250 to 345	246 to 355	1,518	8,697
VGO/RES	C20 to C50	345 to 540+	347 to 595	4,971	31,592
Total	---	---	---	9,969	57,815

THIS TABLE IS FOR ILLUSTRATIVE PURPOSES ONLY AND DOES NOT CORRESPOND TO THE CURRENT DESIGN PROBLEM.

Table 2. Sulfur and water content for distillate cuts from Poseidon Crude. Design values are from CHEMCAD. Light ends, light naphtha, heavy naphtha, and kerosene require further refining.

Product	Sulfur, spec, mg/kg	Sulfur, design, CC	H_2O, spec	H_2O, design
Light Ends / Tail Gas	10-100 [1]	600,000	10-30 [7]	29,501
Light Naphtha	300-400 [2]	10,680	582 [8]	383
Heavy Naphtha	1100 [3]	384	25 [9]	1,886
Kerosene (ultralow to K2)	15 to 3000 [4]	0.485	500-1000 [10]	2,256
GO/D	15 [5]	0.011	500-1000 [11]	853
VGO/RES	10,000 [6]	0.003	1000 [12]	191

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Table 3. Results of economic analysis of PYOIL process.

Capital Investment (millions of dollars)	
Purchased Equipment Cost	4.085
Working Capital, 15% of TCI	3.216
Total Indirect Costs	5.297
Total Direct Costs	12.582
Fixed Capital Investment	17.879
Total Capital Investment	21.095
Measures of Profitability	
Return on Investment, %	19.69
Payback Period, years	3.59
Net Present Worth, million $, at 6% interest	23.336
Net Present Worth, million $, at 12% interest	12.614
Internal Rate of Return, %	24.61
Profitability Benchmarks	
Bond Rate, after taxes, from Table 7-1, %	3.25
Preferred Stock Dividend, from Table 7-1, %	8.00
Common Stock Dividend, from Table 7-1, %	9.00
Minimum Acceptable Return, from Table 8-1, %	12.00
Ref. Payback Period, at MAR, years, eq 8-2c	4.14

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Table 4. List of purchased equipment costs from CHEMCAD for atmospheric crude and light ends units.

Number	Name	Description	CHEMCAD Unit	Cost
100	V-100	Desalter	Multipurpose Flash	$534,516
101	V-101	Lower Column	TOWR Distillation Column	$360,630
102	V-102	Upper Column	TOWR Distillation Column	$290,844
103	V-103	Reflux Drum	Multipurpose Flash	$60,036
104	V-104	Kerosene Stripper	TOWR Distillation Column	$20,954
105	V-105	GO/D Stripper	TOWR Distillation Column	$33,113
106	V-106	Lt Naphtha Recovery	TOWR Distillation Column	$119,574
107	V-107	Lt Naphtha Stabilizer	TOWR Distillation Column	$190,933
201	E-201	Lt Naphtha Cooldown	Heat Exchanger	$4,345
204	E-204	Heavy Naphtha Cooldown	Heat Exchanger	$7,230
205	E-205	Kerosene Cooldown	Heat Exchanger	$8,142
207	E-207	GO/D Cooldown	Heat Exchanger	$11,339
301	E-301	Lt Naphtha Energy Recovery	Heat Exchanger	$29,827
302	E-302	Condenser Energy Recovery	Heat Exchanger	$15,272
303	E-303	Upper Pump-around	Heat Exchanger	$8,488
304	E-304	Heavy Naphtha Energy Recovery	Heat Exchanger	$24,772
305	E-305	Kerosene Energy Recovery	Heat Exchanger	$24,012
306	E-306	Lower Pump-around	Heat Exchanger	$8,226
307	E-307	GO/D Energy Recovery	Heat Exchanger	$111,498
308	H-308	Fired Heater	Fired Heater	$1,136,449
309	E-309	Condenser	Air-Cooled Heat Exchanger	$423,151
401	P-401	Crude Oil Feed	Pump	$9,271
402	P-402	Desalted Crude	Pump	$10,464
403	P-403	Lower Pump-around	Pump	$8,244
404	P-404	Upper Pump-around	Pump	$8,292
405	P-405	Reflux	Pump	$8,627
406	P-406	Light Naphtha	Pump	$9,045
407	P-407	VGO/RES	Pump	$8,401
408	P-408	GO/D	Pump	$9,400
409	P-409	Kerosene	Pump	$8,449
410	P-410	Heavy Naphtha	Pump	$9,103
411	P-411	Lt Naphtha	Pump	$8,816
501	C-501	Lt Ends	Compressor	$563,607

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Table 5. Utility usage rates from CHEMCAD for atmospheric crude and light ends units.

Number	Name	Description	Rate, $/hr
308	H-308	Fired Heater	100.91
309	E-309	Condenser	90.94
401	P-401	Crude Oil Feed Pump	18.10
402	P-402	Desalted Crude Pump	0.28
403	P-403	Lower Pump-around Pump	0.78
404	P-404	Upper Pump-around Pump	0.04
405	P-405	Reflux Pump	0.06
406	P-406	Light Naphtha Pump	0.17
407	P-407	VGO/RES Pump	0.02
408	P-408	GO/D Pump	0.08
409	P-409	Kerosene Pump	0.01
410	P-410	Heavy Naphtha Pump	0.02
411	P-411	Lt Naphtha Pump	0.01
501	C-501	Compressor	18.10

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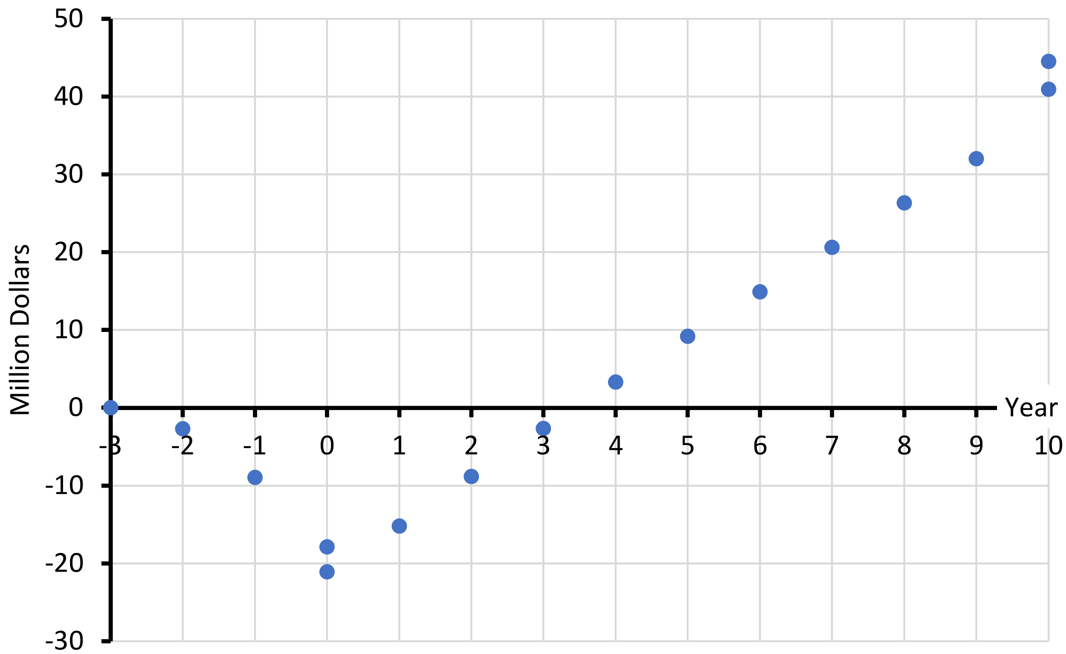

Figure 1. Cumulative cash position for atmospheric crude and light ends units processing 10,000 BPSD of Poseidon crude, as per Figure 6-2 in Peters, Timmerhaus, and West. The payback period (payout time) is 3.59 years.

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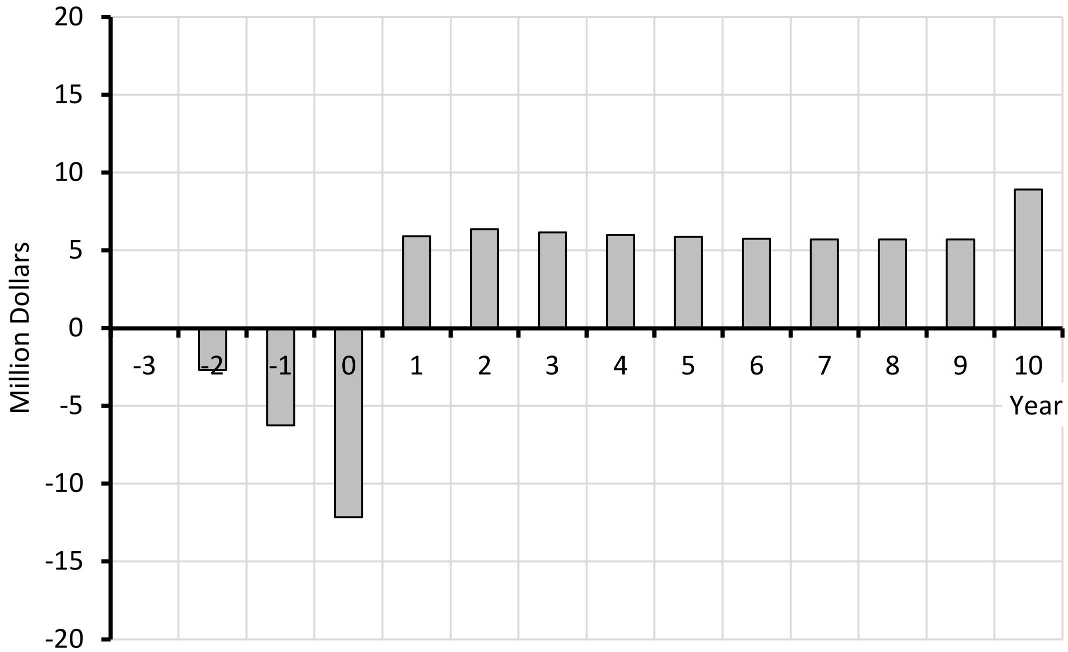

Figure 2. Annual cash flows (from colorful worksheet) for atmospheric crude and light ends units processing 10,000 BPSD of Poseidon crude, as per Figure 7-4 in Peters, Timmerhaus, and West.

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Table 6. Exposure and safety factors for crude oil and distillates.

Species or Fraction	Human Toxicity		Flammability Limits, % in air		Flashpoint, °C	Autoignition Temp., °C	NFPA 704 [6]		
	LD50	LC50	LFL	UFL					
Methane	N/A	500,000 ppm 2h mouse 325 g/m^3 2h mouse [7] 658 mg/L 4h rat [8]	5.3 [9]	14.0 [9]	-188 [4]	537 [9]	2 [9]	4 [9]	3 [9]
Ethane	N/A	658 mg/L 4h rat [10]	3.0 [11]	12.5 [11]	-135 [11]	472 [11]	1 [11]	4 [11]	0 [11]
H_2S	N/A	713 ppm 1h rat [12] 673 ppm 1h mouse [12] 634 ppm 1h mouse [12] 444 ppm 4h rat 4h [12] 700 mg/m^3 35 min rhesus monkey [13]	4.0 [13]	44.0 [13]	-82.4 [12]	260 [13] 232 [12]	4 [13]	4 [13]	0 [13]
LPG	N/A	> 30 mg/L 4h rat [14]	1.8 [15]	13.0 [15]	-5 [15]	400-450 [15]	2 [15]	4 [15]	0 [15]
Naphtha	>5,000 mg/kg rat (oral) [16] >2,000mg/kg rabbit (dermal) [16]	>5 mg/L 4h rat [16]	1.1 [17]	5.9 [17]	37.8 [16]	288 [17]	2 [16]	4 [17]	0 [16]
Gas Oil	>5,000 mg/kg rabbit (dermal) [18] >5,000 mg/kg rabbit (oral) [18]	>6 mg/L 4h rat inhalation [18]	0.6 [18]	6.5 [18]	52 [18] 52-96 [19]	257 [18] 257 [19]	1 [18]	2 [18]	0 [18]
Residuum	>2,000 mg/kg rabbit (dermal) [20] >5,000 mg/kg rat (oral) [20]	>94.4 mg/m^3 5h rat [20]	N/A	N/A	204 [20] 158-275 [21]	485 [20]	0 [20]	3 [20]	0 [20]
Crude Oil	>2,000 mg/kg rabbit (dermal) [22] >4,300 mg/kg rat (oral) [23]	3.95 mg/L [24]	1 [25]	7 [25]	-29 [23] 0-40 [25]	454 [25]	2 [25]	3 [25]	0 [25]

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Table 7. Pressure relief valve sizing from API 520 standard.

		Steam, Main Column	Steam, Kerosene Stripper	Steam, GO/D Stripper	Vapor, Column Overhead	Liquid, Column Bottom
Stream		901	902	903	251	7
Set Pressure	psia	65	65	65	40	40
Set Pressure	kPa	448	448	448	276	276
Relieving Temperature	°C	500	500	500	102	---
Relieving Temperature	K	773	773	773	375	---
Mass Flow Rate	kg/hr	6000	500	500	54656	
Vol Flow Rate	L/m	---	---	---	---	10179
Density	kg/m^3	---	---	---	---	774
Viscosity	cP	---	---	---	---	0.788
Rated Coefficient of Discharge(Kd)	---	0.975	0.975	0.975	0.975	0.65
Backpressure Correction (Kb)	---	1	1	1	1	1
Rupture Disc Correction Factor(Kc)	---	1	1	1	1	1
Cor. Factor(Kn) for Steam Pressure	---	0.945	0.945	0.945	---	---
Cor. Factor for Steam Superheat (Ksh)	---	0.764	0.764	0.764	---	---
Required Effective Discharge Area(A)	mm^2	2735.4	227.9	227.9	11453.8	9431.3
Calculated Diameter	cm	5.9	1.7	1.7	12.1	11.0
Selected Standard Diameter	cm	6.0	2.0	2.0	12.5	11.0
Selected Diameter	in	2.5	0.75	0.75	5.0	4.5
Calculations performed at www.codecalculation.com						

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Table 8. Environmental and ecological assessment of crude oil distillates and hydrogen sulfide. Data are taken from Google AI searches and Safety Data Sheets. Risk to aquatic organisms is considered very high if the EC50 (median effective concentration) is less than about 1 mg/L and high at 1-5 mg/L. Global warming potential of methane and ethane are considered high since these are direct heat-trapping molecules. A score of moderate appears when significant amounts of CO_2 are emitted by combustion of the distillate as fuel.

	EC50, 48h, Daphnia magna, mg/L	EC50, 72h, Algae, mg/L	EC50, 96h, Fish, mg/L	Risk of Persistence & Degradability	Bioaccumulation Potential	Global Warming Potential (GWP)	Global Warming Assessment
Sulfur, as H_2S	.06	1.9	.01-.1	Low	Low	25	High
Methane	69.4	19.4	147.5	Low	Low	25	High
Ethane	46.6	16.5	91.4	Low	Low	6	High
Propane	27.1	11.9	49.9	Low	Low	3	High
i-Butane	16.3	8.6	28.0	Low	Low	3	High
n-Butane	14.2	7.7	28.0	Low	Low	3	High
n-Pentane	2.3	1.3	4.3	Low	Low	5-11	Low
Light Naphtha	.6-14	.6-13	1-10	Low	Low	4	Low
Heavy Naphtha	.6-3	.3-5	.1-10	Low	Mod	4	Low
Kerosene	1.6-16	.5-5	1-5	Mod	High	3	Low
Gas Oil / Diesel	.4	2.5	20	Mod	High	3	Low
Heavy Gas Oil / Residuum	2.0	2.0	21	High	High	.4	Low

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